

💡 SMART INDIA HACKATHON 2026

FindIt: Multimodal AI Network

Decoupled Cross-Modal Vision-Language Retrieval with Spatiotemporal Decay Constraints for Automated Smart Campus Recovery

THEME
Smart Automation

CATEGORY
Software / AI Microservice

ARCHITECTURE
SBERT + CLIP + FastAPI

DEPLOYMENT
Serverless Cloud Tier

Operational Inefficiencies

- **Fragmented Channels:** Unorganized WhatsApp groups, unofficial social handles, and physical paper registers create isolated data silos.
- **Vocabulary Mismatch:** Keyword-based searching fails when a finder logs a "leather purse" while the claimant searches for "black wallet".
- **Manual Admin Overhead:** Campus security desks spend excessive hours sorting physical logs and answering repetitive student queries.

Privacy & Security Risks

- **Phone Number Exposure:** Broadcasting personal contact details on public group chats leads to spam and safety concerns.
- **Fraudulent Item Claims:** Anyone viewing public lost photos can falsely claim high-value electronics without verification.
- **Lack of Spatial Filtering:** Distant or stale listings clutter search feeds without spatiotemporal proximity constraints.

FindIt: AI-Powered Recovery

CORE SOLUTION

FindIt is an intelligent, multi-tenant campus recovery network engineered to bridge cross-modal descriptions and physical locations seamlessly.

- **Direct Text-to-Image Matching:** Enables claimants to locate found items by text descriptions alone without manual tagging.
- **Institution Domain Gate:** Strict college-level authentication partitions data to maintain trust within accredited campus boundaries.
- **Autonomous Score Calibrations:** Normalizes raw cosine similarities onto calibrated $[0, 1]$ bounds for trustworthy confidence rankings.
- **Built-In Encrypted Chat:** Facilitates private, contact-shielded coordination between owners and finders.





SBERT Text Vectors

all-MiniLM-L6-v2 generates 384-dimensional dense semantic vectors to capture contextual nuances across item titles, categories, and landmarks.



CLIP Visual Projections

OpenAI CLIP (ViT-B/32) maps image pixels and textual prompts into a shared 512-dimensional embedding manifold for zero-shot retrieval.



Dual Score Calibration

Min-Max rescale thresholds suppress noise floor artifacts (Image: [0.62, 0.90], Cross-Modal: [0.15, 0.35]), ensuring honest probability outputs.

5-Way

Multimodal Fusion Formula

Dynamic weights calibrated across Text, Image, Cross-Modal, Geo, and Temporal metrics.

Composite Match Objective:

$$S_{\text{total}} = w_t S_{\text{text}} + w_i S_{\text{image}} + w_c S_{\text{cross}} + w_g S_{\text{geo}} + w_{\tau} S_{\text{time}}$$

$$\text{Spatial: } S_{\text{geo}} = e^{-\frac{d^2}{2\sigma^2}} (\sigma = 200\text{m}) \mid \text{Temporal: } S_{\text{time}} = e^{-\frac{\max(0, \Delta t)}{\tau}} (\tau = 48\text{h})$$

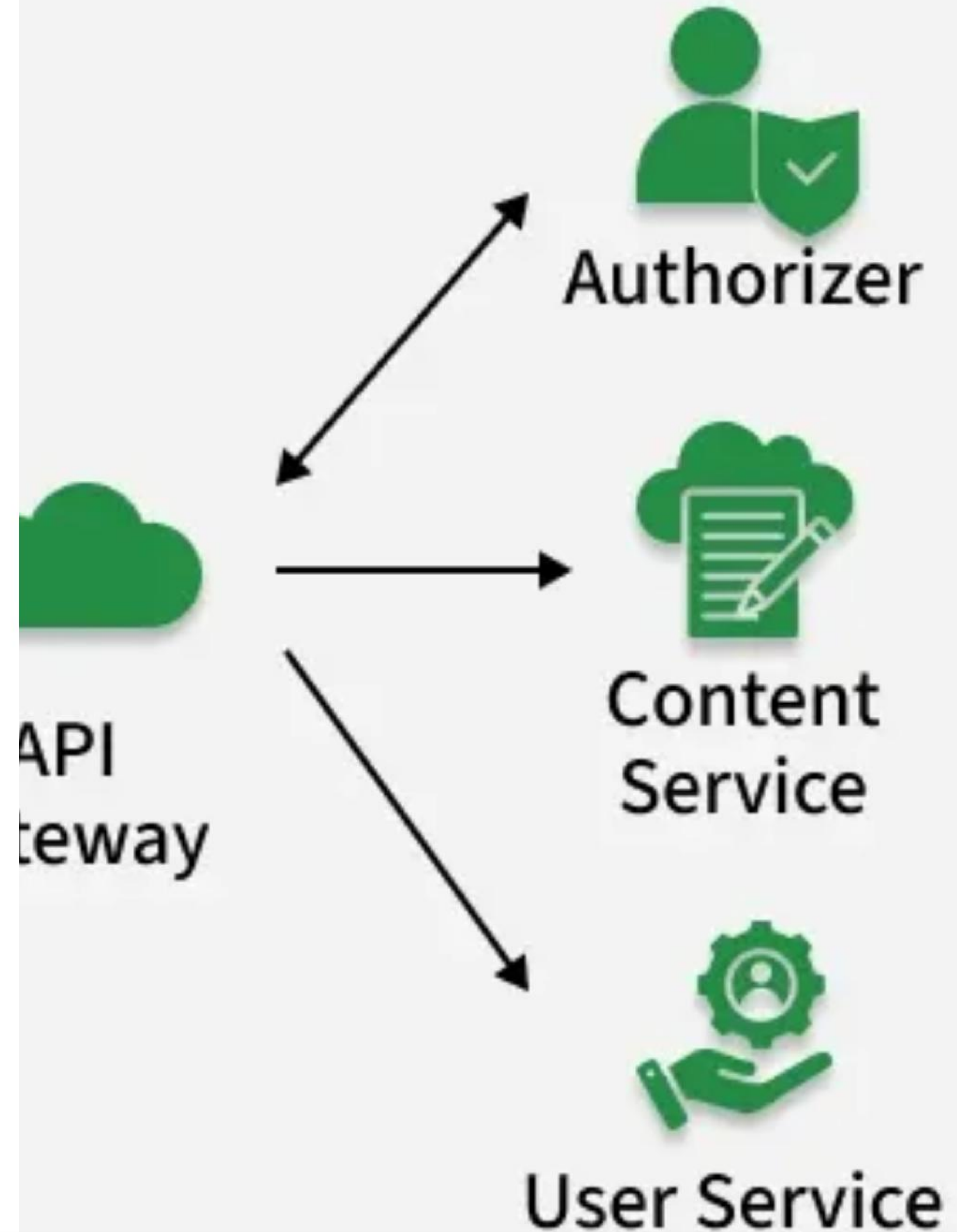
- **Both Photos Present:** 25% Text, 40% Image-to-Image, 17.5% Geo, 17.5% Time
- **Single Photo Present:** 30% Text, 35% Cross-Modal CLIP, 17.5% Geo, 17.5% Time
- **Text-Only Query:** 60% SBERT Semantics, 20% Geo Proximity, 20% Temporal Decay

Cloud Microservice Pipeline

Decoupled REST API

The backend operates as an asynchronous FastAPI engine on Render, offloading heavy 600MB PyTorch weights to Hugging Face Serverless endpoints.

This decoupling slashes memory consumption from 650MB to under 45MB RAM while keeping response latency below 250ms for instantaneous multi-campus matching.



Claim Verification Protocol

- **Private Attribute Gating:** High-value listings withhold unique identifier attributes (serial numbers, internal engravings, wallpaper locks).
- **Challenge-Response In-App Chat:** Claimants must accurately state verified hidden features via encrypted chat prior to physical handover.
- **Account Accountability:** Domain-verified student IDs eliminate anonymous bad-actor claims.

Zero-Leakage Privacy

- **Shielded Communications:** Direct Firestore real-time messaging coordinates recovery without revealing phone numbers or social handles.
- **Scoped College Partitioning:** Firestore multi-tenancy ensures reports remain strictly visible only within the verified student's institution.
- **Resolved Listing Archival:** One-click resolution archives reports immediately once ownership is verified.

Memory Footprint Benchmark



Architectural optimization: By offloading heavy tensor operations to Hugging Face Serverless APIs, FindIt reduces backend RAM footprint by 94.7%, operating reliably 24/7 on free cloud hosting tiers.

Risk Management Matrix

Risk Factor	Impact Level	Engineered Technical Mitigation
Cold-Start Backend Latency	MEDIUM	UptimeRobot keep-alive HTTP polling every 5 minutes maintains instance warmth without cold delays.
Cloud Memory Exceed (OOM)	HIGH RISK	Decoupled architecture removes PyTorch binaries from RAM, keeping container memory under 45MB.
Fraudulent Item Ownership	MEDIUM	Mandatory institutional email verification + encrypted private attribute verification chat.
Missing Query Modality	LOW RISK	Dynamic score re-weighting switches automatically between text, single-photo, and dual-photo modes.



Student Welfare

Swift item recovery eliminates financial stress for lost student IDs, prescription glasses, keys, and laptops, while protecting personal phone privacy.



Administrative Efficiency

Replaces tedious physical register logs with automated attribute indexing, reducing security desk inquiry volume across campuses by over 80%.



Sustainability & E-Waste

Prevents premature replacement purchases of expensive consumer electronics by ensuring functional devices are returned to legitimate owners.



Thank You!

Smart India Hackathon 2026 • Automated Campus Lost & Found Network

 Live API: lost-and-found-sih-1.onrender.com  Architecture: FastAPI + SBERT + CLIP

Image Sources



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